SECTION A **[40 MARKS]**

Read the case study below and answer ALL the questions that follow.

PORT CONGESTION BOGS DOWN SOUTH AFRICA'S ECONOMY

Port congestion marked by swarms of anchored ships is a distant pandemic memory in most global trade hubs, boosting expectations for the first disruption-free period for holiday shipping season since 2019. Satellite data currently show nearly almost 100 cargo vessels — carrying fuel, bulk dry goods, containers and cars — are waiting outside the ports of Durban and Richards Bay, including some anchored near Cape Town and Gqeberha. The terminals are dealing with issues ranging from bad weather to aging equipment. According to the latest note from German container carrier Hapag-Lloyd, there's a six-day delay for berth space in Durban. The backlog in part prompted Maersk to bypass Cape Town as a stop on one of its Asia routes.

Such snarls are frustrating for companies to plan around. For some industries where consumer demand patterns change rapidly, it's more than a headache. Woolworths, for one, has been struggling for some years to get its fashion mix in South Africa right. The battle to turn that unit around is being made all the more difficult by the port bottlenecks. It's costly for everyone involved. MSC, the world's biggest container line, will apply a congestion surcharge of $210 per 20-foot container for dry cargo through South African ports starting Dec. 3. The South African Association of Freight Forwarders, in an emailed statement on Tuesday, said delays have direct costs of 98 million rand ($5.2 million) a day when congestion surcharges are added.

Transnet, the state-run logistics company, said it's undertaking several urgent interventions to address the backlogs and minimize the impact on the South African economy. In a statement this week, Transnet acknowledged the fixes won't be quick after "many years of underinvestment in equipment and its maintenance." "We are working on a number of measures to turn the situation around," Board Chairperson Andile Sangqu said at a media briefing. "We need to caution that this is going to take some time as the lead times for some of the equipment is anything from 12 to 18 months." Logistical challenges including the long delays at ports, frequent power cuts, roads in poor condition, crime and corruption remain pressing concerns for South African businesses, according to data compiled by FirstRand's Rand Merchant Bank unit and Stellenbosch University's Bureau for Economic Research.

The South African Reserve Bank's monetary policy committee has raised interest rates by 475 basis points to 8.25% in its current tightening cycle that started in November 2021. It held rates steady at its last two meetings, and the next rate announcement will be on Thursday. South African annual inflation quickened to a five-month high in October, Pretoria-based Statistics South Africa said Wednesday in a statement on its website. That's unlikely to persuade the central bank to hike rates because the acceleration should be temporary.

Extracted from: https://www.bloomberg.com/news/newsletters/2023-11-22/supply-chain-latest-south-africa-s-port-congestion

QUESTION 1 **[40 MARKS]**

1.1 Considering the condition of ports in South Africa, analyse the potential impacts of the highlighted issues on the operations of retailers like Woolworths. Additionally, propose strategies that Woolworths should contemplate to mitigate these challenges. Support your suggestions with relevant examples. **(20 Marks)**

1.2 In a statement this week, Transnet acknowledged the fixes won't be as quick. "We are working on a number of measures to turn the situation around," Board Chairperson Andile Sangqu said at a media briefing. As a competent operations practitioner demonstrate the various measures and strategies that you would present for adoption to the Board Chairperson to improve Transnet's operations capabilities. **(20 Marks)**

SECTION B **[60 MARKS]**

Answer ANY THREE (3) questions in this section.

QUESTION 2 **(20 Marks)**

Nestlé, a multinational food and beverage corporation headquartered in Vevey, Switzerland, manufactures a diverse array of beverages, encompassing coffee (Nescafé), tea (Nestea), bottled water (Nestlé Pure Life), and dairy-based drinks (Nesquik). The company is exploring the possibility of establishing a manufacturing facility in Southern Africa, with potential locations being Lesotho or Swaziland. You have been tasked with delivering a thorough report outlining the factors that should be taken into account when making this decision. Refer to pertinent content in your analysis.

QUESTION 3 **(20 Marks)**

The Coca-Cola Company operates several manufacturing plants in South Africa, where it produces a wide range of beverages to meet the demands of the local market. At these manufacturing facilities, Coca-Cola produces various beverage products, including carbonated soft drinks, juices, bottled water, energy drinks, and sports drinks. These products are manufactured using state-of-the-art production technologies and adhere to stringent quality and safety standards. In one of its plants in Bloemfontein, it needs to make 100,000 litres of grapetiser per month to meet demand. Each month contains 20 working days, each of which allows for three (3) 8-hour shifts. Within a close radius, the organisation operates a Coke Zero processing unit which has 100,000 labor hours available for the quarter and currently has a labor productivity of 5 gallons per 4,000 hours.
Using the provided information determine the following:

3.1 For the grapetiser processing plant, if a worker can produce 10 litres/hour, determine the number of workers needed on each shift. Suppose each shift has 100 workers, what is the productivity of an individual worker? Lastly, if material costs are R10/litre, capital costs are R100,000 and labor costs are R10/hour, what is the multifactor productivity of this processing plant? **(14 Marks)**

3.2 For the coke-zero processing unit, determine the amount of gallons that the processing plant can make this quarter. If the multi-factor productivity was 1 gallon per R25000, how much should the organisation spend this quarter in manufacturing Coke Zero gallons? **(6 Marks)**

QUESTION 4 **(20 Marks)**

Ezemvelo Muskers purchases small, imported trinkets in bulk, packages them, and sells them to retail stores. The managers are conducting an inventory control study of all their items. The following data for 2023 is provided for item, which is not seasonal.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sales	51	55	54	57	50	68	66	59	67	69	75	77

Using relevant calculations, determine the forecasts for the first FOUR (4) months of 2024 (January, February, March, and April) using trend projection.

QUESTION 5 **(20 Marks)**

5.1 Moses Mabhida Stadium is situated at the heart of KwaZulu Natal's economic hub. It has a conference center that is capable of handling 1,500 participants. However, conference management personnel believe that only 1,300 participants can be handled effectively for most events. The last music event hosted, although forecasted to have 1,300 participants, resulted in the attendance of only 1080 participants. Determine efficiency and utilisation for the conference facility. **(4 Marks)**

5.2 Checkers Hyper in Rosebank sells 175 units per month of OMO 5kg washing powder. The unit cost of this product to the store is R2.50 and the cost of placing an order has been estimated to be R12.00. Checkers Hyper uses an inventory carrying charge of I = 27% per year. Provided this information, calculate (a) the optimal order quantity, (b) the order frequency, and (c) the annual holding and setup cost. If, through automation of the purchasing process, the ordering cost can be cut to R4.00, what will be (d) the new economic order quantity, (e) the order frequency, and (f) annual holding and setup costs? Explain these results. **(16 Marks)**

SECTION A **[40 Marks]**

Read the article below and answer ALL the questions that follow.

The need for speed

The Foschini Retail Group (TFG), one of South Africa's leading fashion and lifestyle retail chains, consists of 14 trading formats dealing in products that range from fashion, jewellery, cosmetics, sporting and outdoor apparel and equipment to homewares. The group, which was formed in 1925 by George Rosenthal, a Russian immigrant, trades in over 1,650 stores, making it the foremost specialty retailer in South Africa.

Today the business employs in excess of 15,000 staff and processes more than 40 million units of stock per year across its six distribution centres. The company has enjoyed a period of steady growth and the existing legacy system was struggling to cope with the sharp peaks in customer demand and the need to adapt to change in a short space of time. For example, each time an adjustment was made to a warehouse process, the existing system would require program development which could take several months.

"It can be easy to identify areas where you can make improvements in a DC environment, but if those changes take months to implement then you are losing out," explained Jan Tukker, group logistics director, TFG. "We needed to upgrade to a system that offered us much more flexibility and could provide a template for a quick, effective roll-out across the remaining divisions."

Learning from Industry leaders

TFG's supply chain team realised that it had built systems and processes that enabled it to operate extremely efficiently. The challenge was that this efficiency didn't fully incorporate the dynamic needs of the trading divisions that it serviced and was related to one static way of operating. The approach taken was to gather as much information as possible and learn from best in class companies. "Rather than fight against the tide, we wanted to understand why our peers operated in a certain manner and how this made them leaders in their industry" explained Jan Tukker. "After all they must be operating in a certain manner and achieving the success that they have for a reason.

Based on this we realised that one of the key components needed to improve our flexibility and capability was a leading Warehouse Management System".

Due Diligence

TFG conducted a thorough review of the market and short-listed three suppliers for consideration. Manhattan and Supply Chain Junction, a specialist supply chain consultancy focusing on the South African market, proved to be best in class and beat off the competition. "Manhattan's retail and domain expertise made it an obvious choice for us," Jan Tukker explains. "The company is not only ranked highly by independent analysts, but has high calibre customer testimonies that gave us the proof we needed. We wanted a partner that we could learn from, and we also appreciated the opportunity to interact with similar businesses to share experiences. We choose Supply Chain Junction as our local partner as they had previously successfully implemented Manhattan's WMS in South Africa. They had a team of local industry experts encompassing not only the software expertise but strong business process knowledge which enabled us to significantly reduce our implementation cost and flexibility as a result of a local rate structure and resource availability". The implementation at the first site was completed within 12 months, the team have however, produced a system template which can easily be dropped into the remaining sites. The second site was live after just two months with the remaining 3 DCs due for completing during 2011. "The joint collaboration of The Foschini Group, Manhattan and Supply Chain Junction cannot be faulted," commented Jan Tukker. "The warehouse management system was delivered on time and under budget thanks to the commitment of the team. We have just completed the implementation at highest volume DC which accounts for approximately 40% of the group's revenue. This was our third go live and during this implementation we also upgraded to the latest version of the software. On the day we went live we shipped just under our full plan and within 2 days we were shipping in excess of our store distribution plan."

The Benefits

"We have met one of our main objectives in that we have designed a series of system templates which can be dropped into the remaining sites. Whereas it may previously have taken us up to six months to deploy cross docking with our legacy system, we can now do the same thing in matter of days," explained Jan Tukker. "Another key benefit has been the improved receiving process which has enabled not only ourselves but our suppliers to plan ahead with Advanced Shipment Notifications, thus enhancing our inbound reliability and accuracy and setting the foundation for improved cross-docking, faster DC throughput and fresher stock on our store shelves." "As a result of the Manhattan system deployment and the process improvements we have made, we've been able to reduce logistics lead times from 17 to 4.9 days in the distribution centres; a DC stockholding reduction of 64 percent; an overall delivery conformance improvement of 11.5 percent and we've measured DC pick accuracy consistently at 99.9 percent. This and the other logistics enhancement initiatives we've undertaken have delivered a 10.1 percent reduction in the overall cost of logistics as a percentage of turnover, and this at a time when cost inflation has been running at around 6 percent." Jan Tukker, TFG.

Tukker continued, "Four years ago, TFG's Board identified supply chain improvement as a top priority. The improvements we have been able to achieve are testament to the hard work of our logistics team, to the strength of processes we have put into place, and to the quality of the people and technologies at our partners including Manhattan Associates and Supply Chain Junction. These processes, people and technologies have come together and enabled us to pursue a path of continuous innovation and improvement to meet the Board's objective."

TFG's phased deployment will see all six DCs running on Manhattan's software across the 14 trading formats. "Manhattan's warehouse management system is already providing the foundation for Foschini's supply chain transformation," concluded Jan Tukker. "We are looking forward to developing our partnership and continuing to learn from and question our peers whilst being able to adapt quickly as we grow."

Source: www.scjunction.co.za

Answer ALL the questions in this section.

QUESTION 1 **(20 Marks)**

As an employee of the Foschini Retail Group (TFG), appraise the five (5) basic performance objectives (cost, speed, dependability, flexibility and quality) in operations.

QUESTION 2 **(20 Marks)**

Critically analyse how the Foschini Retail Group (TFG) used the objectives in Question 1 above to their advantage to improve their operations.

SECTION B **[60 Marks]**

Answer ANY THREE (3) questions in this section.

QUESTION 3

A company manufactures shoes. It requires 5000 metres of leather per month. The company buys the leather from a supplier at a cost of R150 per metre. The company's inventory carrying cost is estimated to be 5% of the metre cost of leather and the ordering cost is R70 per order.

3.1 Calculate the EOQ. **(6 marks)**

3.2 What is the number of orders per year? **(4 marks)**

3.3 Compute the total cost. **(4 marks)**

3.4 Provide an evaluation on the assumptions on which the Economic order quantity is based. **(6 marks)**

QUESTION 4

You are provided with the following information regarding the sales data of a Manufacturing company.

Month	Sales
January	11
February	14
March	16
April	10
May	15
June	17
July	11
August	14
September	17
October	12
November	14
December	16
January	11
February	

4.1 Calculate the 3 month weighted moving averages and deviations. **(15 marks)**

4.2 Calculate the Mean absolute deviation (MAD) for the Manufacturing company. **(5 marks)**

QUESTION 5 **[20 Marks]**

Bailey et al. (2015), discusses several problems in relation to international purchasing. A French Company, AVO, wants to buy ostrich eggs from an ostrich farm in Oudtshoorn, South Africa. You are required to discuss the relevancy of Bailey's 2015 work regarding this business transaction. (Hint: in your answer take into consideration the country of origin and the country where business is going to done).

QUESTION 6 **[20 Marks]**

Using an appropriate diagram, critically evaluate the different relationships within your organisation of choice. In your answer provide the interelationships and the challenges that one may face, operationally.

Formula Sheet:

$$\bar{x} = \frac{\Sigma x}{n}$$

$$\bar{y} = \frac{\Sigma y}{n}$$

$$\text{Moving average} = \frac{\Sigma \text{ demand in previous } n \text{ periods}}{n}$$

$$\text{Weighted moving average} = \frac{\Sigma((\text{weight for period } n)(\text{demand in period } n))}{\Sigma \text{ weights}}$$

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

$$\text{Forecast error} = A_t - F_t$$

$$MAD = \frac{\Sigma\,|\text{Actual} - \text{Forecast}|}{n}$$

$$MSE = \frac{\Sigma(\text{Forecast errors})^2}{n}$$

$$MAPE = \sum_{i=1}^{n} 100 \frac{|\text{Actual}_i - \text{Forecast}_i| / \text{Actual}_i}{n}$$

$$F_t = \alpha(A_{t-1}) + (1-\alpha)(F_{t-1} + T_{t-1})$$

$$T_t = \beta(F_t - F_{t-1}) + (1-\beta)T_{t-1}$$

$$FIT_t = F_t + T_t$$

$$y = a + bx$$

$$b = \frac{\Sigma xy - n\overline{xy}}{\Sigma x^2 - n\bar{x}^2}$$

$$a = \bar{y} - b\bar{x}$$

$$S_{xy} = \sqrt{\frac{\Sigma y^2 - a\Sigma y - b\Sigma xy}{n-2}}$$

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$

$$y = a + b_1 x_1 + b_2 x_2$$

Answer ALL questions in this section.

Read the two adaptations relating to Apple's 2017 year of operations and answer the questions that follow.

Could Apple's next iPhones be made in the USA?
Apple supplier Foxconn reportedly mulling $7 billion investment in U.S. manufacturing
By Kyle Wiggers — Posted on January 23, 2017

Restoring manufacturing jobs to the United States' struggling Rust Belt communities was one of President-elect Donald Trump's biggest campaign promises — and Apple is stepping up to the plate. The consumer electronics giant is exploring the possibility of moving smartphone production to the United States.

Electronics maker Foxconn, one of Apple's largest suppliers, confirmed on Sunday that it was mulling a $7 billion investment to create a flat-panel manufacturing facility in the United States, Reuters reported. This would bring one of the major components in smartphones to American shores and would be an important step toward building iPhones in the U.S. Founder and chairman Terry Gou said the move may create as many as 50,000 jobs and would involve Japanese subsidiary Sharp; talks were reportedly underway in Pennsylvania and in other states.

Rumors swirl about "Made in the USA"

Speculation on Apple's plans began in late 2016, and heightened following an interview with Donald Trump in The New York Times, during which he recounted a phone conversation with Tim Cook urging the CEO to move part of Apple's production line to the U.S.:

"I was honored yesterday, I got a call from Bill Gates, great call, we had a great conversation, I got a call from Tim Cook at Apple, and I said, 'Tim, you know one of the things that will be a real achievement for me is when I get Apple to build a big plant in the United States, or many big plants in the United States, where instead of going to China, and going to Vietnam, and going to the places that you go to, you're making your product right here.' He said, 'I understand that.' I said: 'I think we'll create the incentives for you, and I think you're going to do it. We're going for a very large tax cut for corporations, which you'll be happy about.'"

Trump has spoken on a number of occasions since about Apple moving production to the U.S. Days before his inauguration, the president-elect spoke with Axios, saying that Cook had his "eyes open to it" and that he thinks Cook "loves this country, and I think he'd like to do something major here."

Such a move may become more feasible given Foxconn's plans. The company first confirmed that it was exploring investing in the U.S. in early December: "We are in preliminary discussions regarding a potential investment that would represent an expansion of our current U.S. operations," Foxconn said to CNNMoney.

Softbank CEO Masayoshi Son met with Trump shortly after to announce a planned $50 billion investment in U.S. represent an expansion of our current U.S. operations," Foxconn said to CNNMoney.

Softbank CEO Masayoshi Son met with Trump shortly after to announce a planned $50 billion investment in U.S. startups. The CEO held a paper with Softbank's and Foxconn's name, along with the following text: "commit to: Invest $50bn + $7bn in US, generate 50k + 50k new jobs in US in next 4 years." That led to speculation that Foxconn would have a role in bringing jobs to the U.S.

"While the scope of the potential investment has not been determined, we will announce the details of any plans following the completion of direct discussions between our leadership and the relevant U.S. officials," the manufacturer told CNNMoney.

Trump is a vocal supporter of U.S. companies that build their products in the U.S. and has proposed levying steep tariffs — potentially as high as 45 percent — on competing Chinese importers.

Nikkei, citing a source familiar with Apple's plans, reports that the Cupertino, California-based company has tasked Foxconn and Pegatron, the two tech firms responsible for assembling more than 200 million of Apple's iPhones annually, with investigating the feasibility of building plants in the U.S.

“We’re going to get Apple to build their damn computers and things in this country instead of other countries,” Trump said in a speech in January. “How does it help us when they make it in China?”

Pegatron reportedly demurred, citing logistical concerns. Foxconn agreed to compile a report as soon as June, but company chairman Terry Gou warned that it would show drastically higher productions costs. The potential result? An iPhone made in the U.S. could retail for as much as $740 to $1,300 for a 32GB iPhone 7 versus $650 today, according to Nikkei.

Apple has previously declined to move iPhones production stateside, citing costs.

What would a U.S.-made iPhone cost?

A thorough report in the MIT Technology Review found that moving iPhone assembly to the U.S. would add $30 to $40 to the cost of an iPhone thanks to “transportation and logistics expenses [that] would arise from shipping parts.”

Manufacturing the smartphone’s hundreds of components domestically is an even pricier — and vastly more complex — proposition. Apple Chief Executive Tim Cook told CBS’ 60 Minutes in December 2015 that the U.S. labor pool lacked the skills necessary to carry out iPhone production, and Apple executives have estimated that it would take as long as nine months to recruit the roughly 8,700 industrial engineers that oversee Chinese assembly lines. And that’s before efficiency is taken into account: A 2012 CNN Money report noted that Chinese factories house workers in employee dormitories and “can send hundreds of thousands to the assembly lines at a moment’s notice.”

Then there’s the U.S.’s lack of natural resources to consider. MIT Technology Review points out that few of the 75 elements required to manufacture the iPhone are available commercially in the U.S. Aluminum, for instance, requires bauxite, and there are no bauxite mines in the U.S. China, on the other hand, produces 85 percent of the world’s rare earth metals.

Further complicating matters is Apple’s sprawling supply chain of more than 750 firms in over 20 countries. Taiwan Semiconductor produces crucial iPhone chips; South Korea’s SK Hynix and Japan’s Toshiba produce the handset’s memory modules, and Japan’s Japan Display and Sharp provide the iPhone’s display. “To make iPhones, there will need to be a cluster of suppliers in the same place, which the U.S. does not have at the moment,” an industry executive familiar with iPhone production told economics blog NorthCrane.

But Apple’s plan isn’t without precedent. In 2013, Motorola Mobility employed more than 3,800 employees to assemble the Moto X, a flagship Android phone, at a factory in Fort Worth, Texas. Just a year later, though, it was forced to shutter production as a result of “exceptionally tough” market conditions, according to Motorola president Rick Osterloh. The company subsequently moved production to China.

Others have been more successful. Foxconn established a stateside iMac computer assembly line in 2012. A year later, Singapore-based Flextronics, the manufacturer of Apple’s Mac Pro desktop computer, built a production line in Austin, Texas.

In October 2015, Sharp president Tai Jeng-wu suggested that if Apple were to begin producing smartphones in the United States, it would likely follow suit. “We are now building a new [advanced organic light-emitting diode] facility in Japan. We can make [OLED panels] in the U.S. too,” he said. “If our key customer demands us to manufacture in the U.S., is it possible for us not to do so?”

Adapted from: https://www.digitaltrends.com/mobile/apple-iphone-us-manufacturing/

QUESTION 1 **(40 Marks)**

By applying Porter’s methods on comparative advantage, conduct a detailed analysis of how the decision to relocate will affect Apple’s operational competitive advantage.

QUESTION 2 **(20 Marks)**

A company makes manufactures shoes. It produces 1 000 pairs of shoes per month. It buys the leather and components from a supplier at a cost of R20 per component kit. The company's inventory carrying cost is estimated to be 15% of the purchasing cost per tyre and the ordering is R50 per component kit.

You are required to calculate:

2.1. Calculate the EOQ. (6 marks)
2.2. What is the number of orders per year? (4 marks)
2.3. Compute the average annual ordering costs. (3 marks)
2.4. Compute the average inventory. (2 marks)
2.5. What is the average annual carrying cost? (2 marks)
2.6. Compute the total cost. (3 marks)

QUESTION 3 **(20 Marks)**

The product life cycle goes through many phases, involves many professional disciplines, and requires many skills, tools and processes. Product life cycle (PLC) has to do with the life of a product in the market with respect to business/commercial costs and sales measures.

Choose a product of your choice, and provide a detailed related analyses relating to the stages of the product life cycle.

QUESTION 3 **(20 Marks)**

The product life cycle goes through many phases, involves many professional disciplines, and requires many skills, tools and processes. Product life cycle (PLC) has to do with the life of a product in the market with respect to business/commercial costs and sales measures.

Choose a product of your choice, and provide a detailed related analyses relating to the stages of the product life cycle.

QUESTION 4 **(20 Marks)**

Use the information in the table below and answer the questions that follow:

ACTIVITY	PREDECESSOR	TIME (Days)
A	-	10
B	-	23
C	-	16
D	A	14
E	B	15
F	C,E	13
G	D	15
H	F	9
I	G,H	4

4.1. Draw the Activity-on-Node diagram for the above information. (15 marks)

4.2 Calculate the duration of all possible paths on the network diagram. (3 marks)

4.3 Identify the critical path and state its duration. (2 marks)

QUESTION 5 **(20 marks)**

ABC Analysis classifies inventory into three categories according to some measure of importance with the intention of allocating control efforts accordingly. The idea is to establish inventory policies that focus resources on a few critical inventory parts and not the many trivial ones. It is not realistic to monitor inexpensive stock with the same intensity as very expensive items.

You are provided with the following information. Conduct an ABC analysis, categorising each item of stock appropriately. (80% = A, 15% = B, 5%=C)

Stock Units	Average demand per unit
A1	440 840
A2	543 289
A3	79 800
A4	45 567
A5	123 098
A6	47 345
A7	645 000
A8	587 908
A9	208 006
A10	13 895